BIOINFORMATICS PROJECT REPORT

Comparative Analysis of Hemoglobin Beta Protein Sequences Across Five Species

Comparative protein sequence analysis using UniProt, Clustal Omega and iTOL

Prepared by: Jayaram Balaji

Course / Institution: Bioinformatics Course, CliniLaunch Research Institute, Bangalore

Date: 06 November 2026

Abstract

This project compares hemoglobin beta protein sequences from five species: human, chimpanzee, mouse, cow and chicken. Sequences were retrieved from UniProt and aligned with Clustal Omega. Pairwise percentage identity was calculated from the multiple sequence alignment, and the Clustal Omega guide tree was visualized using iTOL. Human and chimpanzee showed 100.00% identity in the selected sequences, while cow and chicken showed 63.45% identity. The project demonstrates a basic workflow for protein sequence comparison and visualization.

1. Introduction

Hemoglobin is an important protein found in red blood cells. Its main function is to transport oxygen from the lungs to different parts of the body. Adult hemoglobin contains two alpha-globin chains and two beta-globin chains. This project focuses on the beta-globin protein sequences from five species: human, chimpanzee, mouse, cow and chicken. Bioinformatics tools make it possible to compare protein sequences and examine their similarities and differences. Protein sequences were retrieved from UniProt, aligned using Clustal Omega, and the resulting guide tree was visualized using iTOL. The comparison provides an example of how sequence similarity can be used to explore relationships among species.

2. Aim

To compare hemoglobin beta protein sequences from five species, identify their similarities and differences, and examine the sequence-based relationships shown by the alignment and guide tree.

3. Objectives

1. Retrieve hemoglobin beta protein sequences for five species from UniProt.
2. Use and organize protein sequences in FASTA format.
3. Perform multiple sequence alignment (MSA) using Clustal Omega.
4. Identify conserved and variable amino acid positions in the alignment.
5. Compare sequence identity between each pair of species.
6. Visualize and interpret the Clustal Omega guide tree using iTOL.

4. Materials and Tools Used

Resource / tool	Purpose
UniProt	Retrieve hemoglobin beta protein records and sequences.
FASTA format	Store protein sequences in a standard text format.
Clustal Omega	Perform multiple sequence alignment and produce a guide tree.
iTOL	Display and customize the Newick guide tree.
Computer and text editor	Save, organize and inspect sequence and result files.

5. Methodology

7. Protein records were identified in UniProt using the accession identifiers listed in Table 1.

8. The protein sequences were saved in FASTA format and combined for multiple sequence alignment.
9. The five sequences were submitted to Clustal Omega. The output alignment was downloaded in CLUSTAL format, and the Newick guide-tree output was retained.
10. The Newick tree was uploaded to iTOL and displayed in a rectangular layout.
11. Pairwise percentage identity was calculated from the supplied multiple alignment. For each pair, exact amino-acid matches were divided by the number of alignment columns in which both sequences contained an amino acid; gap-containing pairwise columns were excluded from the denominator.

5.1 Protein records used

Species	UniProt accession	Sequence length (aa)
Human	P68871	147
Chimpanzee	P68873	147
Mouse	P02088	147
Cow (Bovine)	P02070	145
Chicken	P02112	147

6. Results and Discussion

6.1 Multiple sequence alignment

The five protein sequences were aligned with Clustal Omega. The alignment contains conserved positions, variable positions and gaps. In the CLUSTAL consensus line, an asterisk (*) indicates a position with identical residues in all sequences; a colon (:) indicates strongly similar properties, and a period (.) indicates weakly similar properties.

6.2 Pairwise sequence identity

Pairwise identity was calculated for all ten species pairs from the multiple alignment. The denominator excludes alignment columns containing a gap in either member of the pair.

Species pair	Identical positions / compared positions	Identity
Human vs Chimpanzee	147 / 147	100.00%
Human vs Cow	122 / 145	84.14%
Human vs Mouse	118 / 147	80.27%
Human vs Chicken	102 / 147	69.39%
Chimpanzee vs Cow	122 / 145	84.14%
Chimpanzee vs Mouse	118 / 147	80.27%
Chimpanzee vs Chicken	102 / 147	69.39%
Mouse vs Cow	107 / 145	73.79%
Mouse vs Chicken	96 / 147	65.31%
Cow vs Chicken	92 / 145	63.45%

The human and chimpanzee sequences are identical across all 147 aligned positions in this dataset, giving 100.00% identity. Human and bovine show 84.14% identity, while human and mouse show 80.27%. Human and chicken show 69.39% identity. Across all ten pairwise comparisons, cow and chicken have the

lowest identity at 63.45%. These values describe the particular protein sequences and alignment used in this project; they should not be interpreted as whole-genome similarity.

6.3 Phylogenetic guide tree

In the displayed Clustal Omega guide tree, human and chimpanzee form a close pair, and mouse and chicken form another pair. Bovine branches separately from those groupings. This pattern is broadly consistent with the pairwise sequence comparisons, particularly the identical human and chimpanzee sequences. The tree is the guide tree produced as part of the sequence-alignment workflow; it is useful for visualizing sequence relationships, but it is not by itself a fully validated evolutionary phylogeny.



Figure 1. Clustal Omega guide tree for the five hemoglobin beta protein sequences, visualized using iTOL.

7. Conclusion

This project compared hemoglobin beta protein sequences from human, chimpanzee, mouse, cow and chicken. Clustal Omega was used for multiple sequence alignment, pairwise identity values were calculated from the alignment, and iTOL was used to visualize the guide tree. Human and chimpanzee showed the highest identity (100.00%), while cow and chicken showed the lowest identity (63.45%) among the comparisons. The work demonstrates how protein sequence alignment and tree visualization can be used to investigate conserved regions, sequence variation and sequence-based relationships among species.

8. Limitations

- The analysis is based on one selected hemoglobin beta protein sequence per species.
- The displayed tree is a Clustal Omega guide tree and should not be treated as a definitive species phylogeny.
- Pairwise identity percentages depend on the alignment and the stated treatment of gap positions.

9. References

12. UniProt Consortium. UniProt Knowledgebase. https://www.uniprot.org/
13. European Bioinformatics Institute. Clustal Omega. https://www.ebi.ac.uk/jdispatcher/msa/clustalo
14. Interactive Tree Of Life (iTOL). https://itol.embl.de/